

TECHNIQUE MASTERING EXERCISES

D - TRIGONOMETRIC FUNCTIONS

1.

a. $\sin\left(\frac{7\pi}{4}\right) = -\frac{1}{\sqrt{2}}$ b. $\operatorname{cosec}\left(-\frac{7\pi}{4}\right) = \sqrt{2}$ c. $\sin(35\pi) = 0$
d. $\cos\left(-\frac{5\pi}{3}\right) = \frac{1}{2}$ e. $\sec\left(\frac{5\pi}{3}\right) = 2$ f. $\sec\left(\frac{15\pi}{2}\right)$ – not defined
g. $\tan\left(\frac{7\pi}{6}\right) = \frac{1}{\sqrt{3}}$ h. $\cot\left(-\frac{7\pi}{6}\right) = -\sqrt{3}$ i. $\cos\left(-\frac{15\pi}{3}\right) = -1$

2.

a. $\cos \theta = -\frac{24}{25}$ $\tan \theta = -\frac{7}{24}$ $\operatorname{cosec} \theta = \frac{25}{7}$ $\cot \theta = -\frac{24}{7}$
b. $\sin \theta = \frac{3}{5}$ $\tan \theta = -\frac{3}{4}$ $\sec \theta = -\frac{5}{4}$ $\operatorname{cosec} \theta = \frac{5}{3}$
c. $\sin \theta = -\frac{5}{13}$ $\cos \theta = -\frac{12}{13}$ $\cot \theta = \frac{12}{5}$ $\sec \theta = -\frac{13}{12}$
d. $\sin \theta = 0$ $\tan \theta = 0$ $\cot \theta$ - not defined $\operatorname{cosec} \theta$ - not defined